

3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(10);
    }
    else
    {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());

        wait(NULL);
    }

    return 0;
}
```

Output:

```
sushmitha@DESKTOP-6O7RQ7V: ~
sushmitha@DESKTOP-6O7RQ7V:~$ nano fork1.c
sushmitha@DESKTOP-6O7RQ7V:~$ gcc fork1.c -o fork1
sushmitha@DESKTOP-6O7RQ7V:~$ ./fork1
Parent Process
PID : 1190
Child Process
PID : 1191
PPID : 1190
_
```

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    for(i=0;i<3;i++)
    {
        if(fork()==0)
        {
            printf("Child %d PID=%d\n",i+1,getpid());
            return 0;
        }
    }

    for(i=0;i<3;i++)
    {
        wait(NULL);
    }

    printf("All children completed.\n");

    return 0;
}
```

Output:

```
sushmitha@DESKTOP-607RQ7V:~$ nano Multichild.c
sushmitha@DESKTOP-607RQ7V:~$ gcc Multichild.c -o Multichild
sushmitha@DESKTOP-607RQ7V:~$ ./Multichild
Child 1 PID=1285
Child 2 PID=1286
Child 3 PID=1287
All children completed.
sushmitha@DESKTOP-607RQ7V:~$
```